Hyeonbin Park

M.S. Student in Cognitive Science | Biointelligence Lab, Seoul National University

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RESEARCH INTERESTS

My research centers on Vision-Language-Action (VLA) models, Latent Action Models (LAMs) and World Action Models (WAMs) that learn human and robot dynamics directly from data, including unlabeled human video. I am also interested in dexterous hand manipulation.

EDUCATION

Seoul National University **Sep 2026 – Aug 2028 (expected)**
M.S. in Cognitive Science (Advisor: Byoung-Tak Zhang)

Soongsil University **Mar 2020 – Aug 2026**
B.S. in AI Software Engineering

PUBLICATIONS

Conference

[C1] **Back to the Familiar Future: Failure Recovery for VLA Policies via Pre-Imagined Milestone Selection**
Suyeon Shin, Juwon Kim, **Hyeonbin Park**, Hyunseo Kim, Hyundo Lee, Hyung-Sin Kim, Byoung-Tak Zhang
Conference on Robot Learning (CoRL) 2026

Workshop

[W1] **Do VLAs Trained with Latent World Models Benefit from Object Slots?**
Hyeonbin Park, Hee Bin Yoo, Byoung-Tak Zhang
CoRL 2026 Workshop: Do Robots Need World Models?

RESEARCH EXPERIENCE

Graduate Researcher — Biointelligence Lab, Seoul National University **Sep 2026 – Present**
- Developing an object-factored latent action model that learns per-agent dynamics from multi-actor demonstrations, conditioning VLA policies on predicted partner behavior for robot-robot and human-robot collaboration.

Undergraduate Research Intern — Biointelligence Lab, Seoul National University **Sep 2025 – Aug 2026**
- B2FF, a failure-recovery framework that guides a frozen VLA back to pre-imagined milestones, raising success on failure-injected LIBERO from 56.3% to 74.0%
- Collected real-world teleoperation and simulated manipulation data on ALOHA bimanual and UR5 platforms, and fine-tuned robot foundation models (ACT, π0.5, OpenVLA-OFT).
- Developed a 3D object reconstruction and pose estimation pipeline using a Zivid 3D camera.

Undergraduate Research Intern — HUMANE Lab, Soongsil University **Jun 2024 – Aug 2025**
- Extended CFT-CLIP (Finding of ACL 2024) for news thumbnail-article representativeness, raising F1 from 0.815 to 0.892 with a sigmoid-based contrastive objective.

- Benchmarked four counterfactual text generation strategies, showing LLM-generated hard negatives underperform masked-LM ones by up to 8 F1 points.

SELECTED PROJECTS

Tori — Voice-Guided Object Localization for Low-Vision Users Jun 2026

Capstone Design | 3-person team | Role: Camera-beam projector calibration lead

- Built a real-time system that localizes a voice-named object from RGB-D input and guides the user by speech and beam-projected light.
- Implemented Gray-code structured-light camera-projector calibration with sub-pixel homography refinement, reaching 0.26 px stereo RMS.

Stock Analytics Platform — sLM-based investment assistant Jun 2025

Capstone Design | 5-person team | Role: RAG pipeline & sLM evaluation

- Built a RAG pipeline summarizing 48 hours of per-ticker news with a locally served sLM, exposed via FastAPI.
- Benchmarked four open-weight sLMs for summarization and sentiment using GPT-4.1 as judge and Financial Phrasebank as ground truth.

SKILLS

- Languages: Python, C, C++
- Robot Learning: PyTorch, LeRobot, Hugging Face; ACT, π0.5, OpenVLA-OFT
- Robotics & Perception: ROS2, MuJoCo, ALOHA, UR5, Zivid 3D camera, Intel RealSense, Azure Kinect
- Tools: Linux, Git, Weights & Biases

HONORS & AWARDS

Gold Prize (3rd place among 12 finalists), AI Software Competition, Soongsil University 2026

Encouragement Award, 16th Capstone Design Competition, Soongsil University 2026

Merit-based Scholarship, Soongsil University 2025